

Izračun prostorskih indeksov
multispektralnih slik

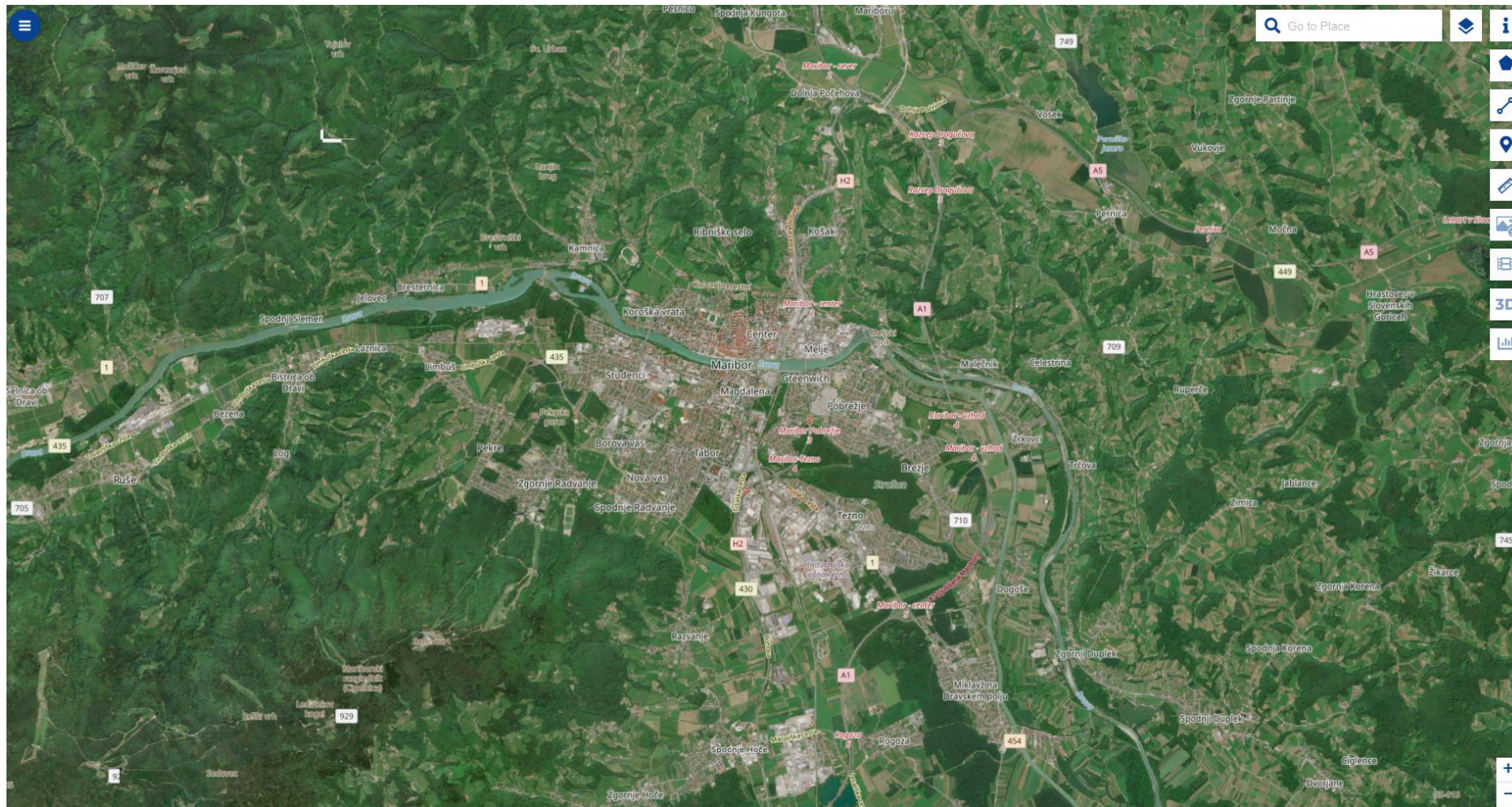
Vsebina

- Pridobivanje podatkov
- Izračun indeksov
- Shranjevanje rezultatov
- Poročilo

Pridobivanje podatkov

- Sentinel 2
 - <https://dataspace.copernicus.eu>
- Na vrhu izberete „COPERNICUS BROWSER“
- Prosto dostopni
 - Potrebna registracija

Pridobivanje podatkov



Pridobivanje podatkov

The screenshot displays the Copernicus Browser interface, which is used for visualizing and searching satellite data. The interface is divided into a left sidebar and a main map area.

Left Sidebar:

- Top Bar:** Includes the Copernicus Browser logo, language selection (EN), a login button, and navigation tabs for "VISUALIZE" and "SEARCH".
- DATA SOURCES:**
 - SENTINEL-1:** Includes a "Filters" link.
 - SENTINEL-2:** Includes a "Filters" link and a sub-selection for "MSI" (checked). Below "MSI" are options for "L1C" and "L2A". A slider indicates a 100% selection.
 - Auxiliary Data File:** Includes a "Filters" link.
 - Immediate:** Includes a "Filters" link.
 - SENTINEL-3:** Includes a "Filters" link.
 - SENTINEL-5P:** Includes a "Filters" link.
 - SENTINEL-6:** Includes a "Filters" link.
 - CCM Optical:** Includes a "Filters" link.
 - CCM DEM:** Includes a "Filters" link.
 - CCM SAR:** Includes a "Filters" link.
 - GLOBAL-MOSAICS:** Includes a "Filters" link.
- TIME RANGE:**
 - From:** Set to "2024-09-10" with time selectors for hours (hh 00) and minutes (mm 00).
 - Until:** Set to "2024-10-09" with time selectors for hours (hh 23) and minutes (mm 59).
 - Filter by months:** A button to filter data by month.
 - Search:** A green button to execute the search.

Main Map Area:

- Displays a satellite image of a region, likely Maribor, Slovenia, showing the Sava River and surrounding urban and rural areas.
- Includes a "Go to Place" search bar at the top right.
- Features a vertical toolbar on the right side with icons for various map functions: information, home, layers, link, location, scale, full screen, 3D view, and a legend.
- At the bottom right, there are zoom in (+) and zoom out (-) buttons.

Pridobivanje podatkov

The screenshot displays the Copernicus Browser interface. On the left, a sidebar contains a search bar and a list of 22 results. Each result includes a thumbnail, a 'Visualize' button, and a 'Download product' button. The main area on the right shows a satellite map of Central Europe, with a large blue rectangle overlaid on the map, indicating a selected area of interest. The interface includes a top navigation bar with 'EN', 'Login', and 'Visualize' buttons. The bottom of the interface features logos for the European Union, Copernicus, and ESA, along with 'About' and 'Support' links.

Search Results:

- S2A_MSIL1C_20241008T095031_N0511_R079_T33TWM_20241008T104310.SAFE**
Mission: SENTINEL-2 Instrument: MSI Size: 740MB
Sensing time: 2024-10-08T09:50:31.024000Z
- S2A_MSIL2A_20241008T095031_N0511_R079_T33TWM_20241008T124449.SAFE**
Mission: SENTINEL-2 Instrument: MSI Size: 1047MB
Sensing time: 2024-10-08T09:50:31.024000Z
- S2B_MSIL2A_20241003T095029_N0511_R079_T33TWM_20241003T125412.SAFE**
Mission: SENTINEL-2 Instrument: MSI Size: 605MB
Sensing time: 2024-10-03T09:50:29.024000Z
- S2B_MSIL1C_20241003T095029_N0511_R079_T33TWM_20241003T120714.SAFE**
Mission: SENTINEL-2 Instrument: MSI Size: 502MB
Sensing time: 2024-10-03T09:50:29.024000Z
- S2A_MSIL1C_20241001T100031_N0511_R122_T33TWM_20241001T105150.SAFE**
Mission: SENTINEL-2 Instrument: MSI Size: 762MB
Sensing time: 2024-10-01T10:00:31.024000Z
- S2A_MSIL2A_20241001T100031_N0511_R122_T33TWM_20241001T123351.SAFE**
Mission: SENTINEL-2 Instrument: MSI Size: 1025MB
Sensing time: 2024-10-01T10:00:31.024000Z
- S2A_MSIL1C_20240928T095031_N0511_R079_T33TWM_20240928T104238.SAFE**
Mission: SENTINEL-2 Instrument: MSI Size: 601MB
Sensing time: 2024-09-28T09:50:31.024000Z
- S2A_MSIL2A_20240928T095031_N0511_R079_T33TWM_20240928T124250.SAFE**
Mission: SENTINEL-2 Instrument: MSI Size: 757MB
Sensing time: 2024-09-28T09:50:31.024000Z

Izračun indeksov

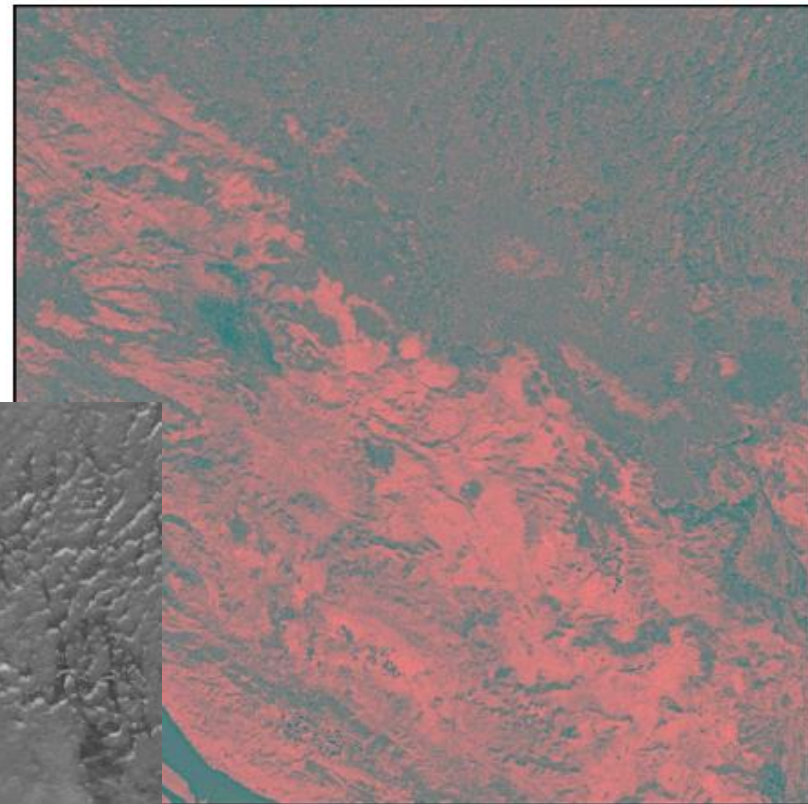
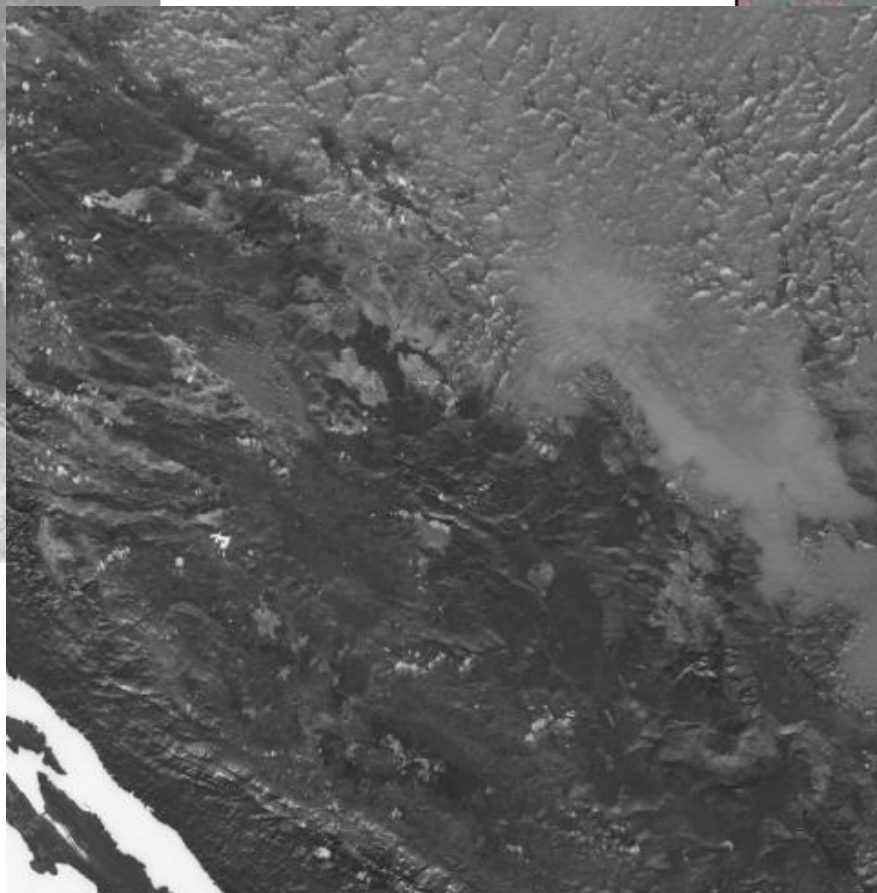
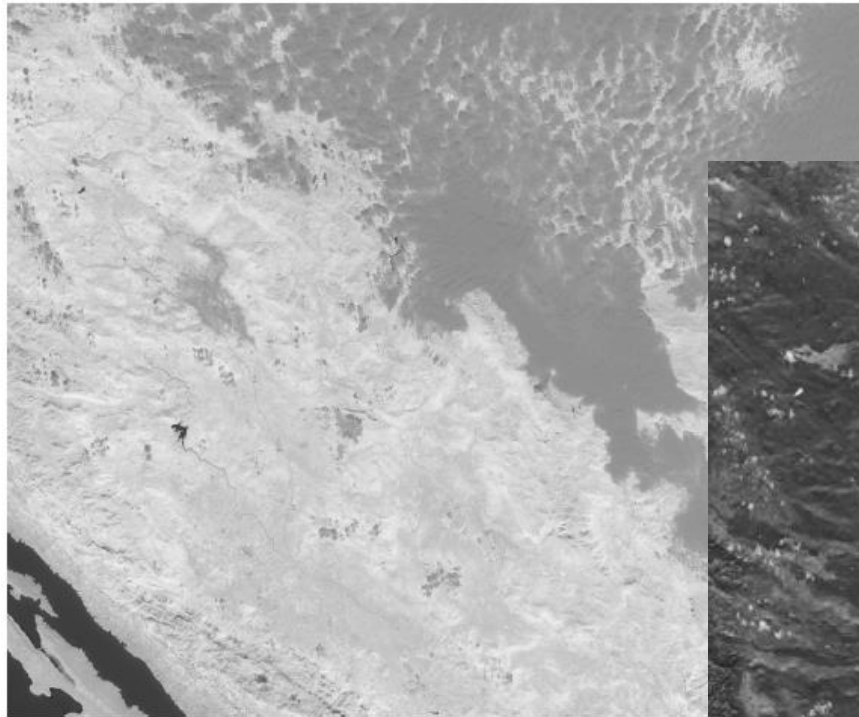
- Posamezen posnetek
 - 13 pasov (angl. band)
 - Podmapa GRANULE/X/IMG_DATA
- Različne ločljivosti!

GRANULE ▶ L1C_T33TWM_A023986_20211009T095128 ▶ IMG_DATA		
Turn	New folder	
Name	Type	Size
 T33TWM_20211009T095029_B01.jp2	IrfanView JP2 File	3.674 KB
 T33TWM_20211009T095029_B02.jp2	IrfanView JP2 File	95.375 KB
 T33TWM_20211009T095029_B03.jp2	IrfanView JP2 File	97.828 KB
 T33TWM_20211009T095029_B04.jp2	IrfanView JP2 File	99.662 KB
 T33TWM_20211009T095029_B05.jp2	IrfanView JP2 File	30.231 KB
 T33TWM_20211009T095029_B06.jp2	IrfanView JP2 File	32.427 KB
 T33TWM_20211009T095029_B07.jp2	IrfanView JP2 File	33.047 KB
 T33TWM_20211009T095029_B08.jp2	IrfanView JP2 File	115.634 KB
 T33TWM_20211009T095029_B8A.jp2	IrfanView JP2 File	33.083 KB
 T33TWM_20211009T095029_B09.jp2	IrfanView JP2 File	3.641 KB
 T33TWM_20211009T095029_B10.jp2	IrfanView JP2 File	1.811 KB
 T33TWM_20211009T095029_B11.jp2	IrfanView JP2 File	31.165 KB
 T33TWM_20211009T095029_B12.jp2	IrfanView JP2 File	30.134 KB

Izračun indeksov

Band name	Resolution [m]	Central wavelength [nm]	Band width [nm]		
1	60	443.9	27		
2	10	496.6	98	Name	Definitions
3	10	560.0	45		
4	10	664.5	38	Enhanced vegetation index	$EVI = \frac{\overline{s} [8] - \overline{s} [4]}{\overline{s} [8] + 6\overline{s} [4] - 7.5\overline{s} [2] + 1}$
5	20	703.9	19		
6	20	740.2	18	Normalised difference vegetation index	$NDVI = \frac{\overline{s} [8] - \overline{s} [4]}{\overline{s} [8] + \overline{s} [4]}$
7	20	782.5	28		
8	10	835.1	145	Green normalised difference vegetation index	$GNDVI = \frac{\overline{s} [8] - \overline{s} [3]}{\overline{s} [8] + \overline{s} [3]}$
8a	20	864.8	33		
9	60	945.0	26		
10	60	1373.5	75	Moisture stress index	$MSI = \frac{\overline{s} [11]}{\overline{s} [8]}$
11	20	1613.7	143		
12	20	2202.4	242	Normalised difference water index	$NDWI = \frac{\overline{s} [3] - \overline{s} [11]}{\overline{s} [3] + \overline{s} [11]}$
				Normalised difference built-up index	$NDBI = \frac{\overline{s} [11] - \overline{s} [8]}{\overline{s} [11] + \overline{s} [8]}$
				Normalised difference mud index	$NDMI = \frac{\overline{s} [9] - \overline{s} [8]}{\overline{s} [9] + \overline{s} [8]}$

Izračun indeksov - primeri



Shranjevanje rezultatov

- Format .tif
 - Poljuben sample format
- Poročilo
 - Indeksi, ki ste jih spoznali na predavanjih
 - 5 poljubnih indeksov

Zaključek

- Vrednost naloge
 - 4% - izračunani indeksi s predavanj
 - 3% - izračunanih 5 poljubnih indeksov
 - 1% - shranjevanje v tif formatu
 - 2% - poročilo
- Ni ustnega zagovora